

04 – Cloud Fundamentals

Module 1 – Cloud Basics

INTRODUCTION :

Earlier, companies had to buy servers, storage, and software to run applications. This was expensive and difficult to maintain.

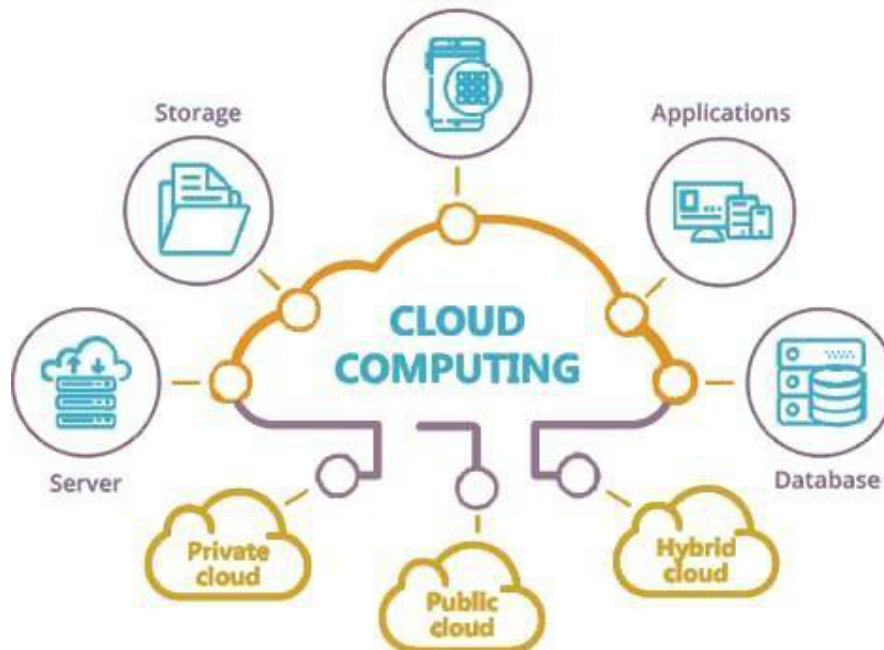
Now, with cloud computing, all these things (storage, servers, apps, software) are available on the internet.

You just pay for what you use - like paying for electricity or mobile recharge.

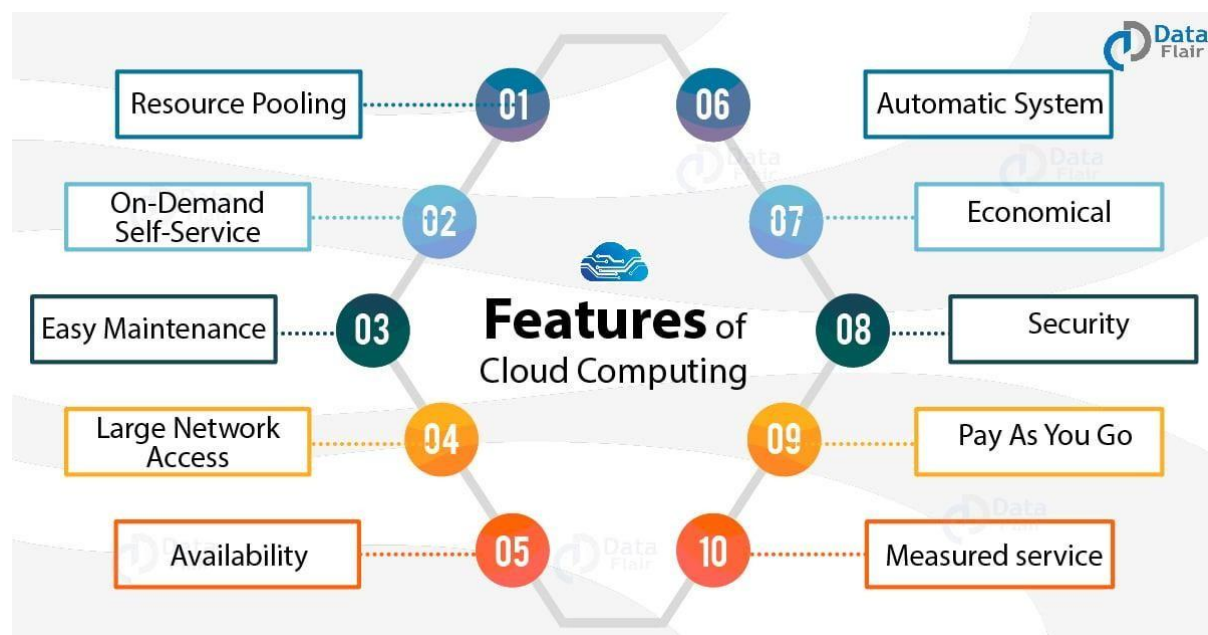
Cloud computing means using the internet to store, manage, and process data instead of using your personal computer or local server.

In simple words, the "cloud" is like a big storage and service system on the internet where companies and people can use resources like servers, storage, applications, and databases without owning them. For example, instead of saving files on your computer, you can save them on Google Drive and access them anywhere. The cloud is flexible (you can use more or less as needed), cost-saving (you pay only for what you use), and reliable (data is backed up and secure). There are three main types of cloud services: IaaS (Infrastructure as a Service), PaaS (Platform as a Service), and SaaS (Software as a Service). Cloud is used in almost every field today-education, business, healthcare, banking, and more.

Cloud providers (like Google, Amazon, Microsoft) have big data centers with powerful computers. When users need resources, they connect to these data centers over the internet. It feels like you are using your own system, but in reality, it is running on remote servers



FEATURES OF CLOUD COMPUTING:



Cloud computing has some key features that make it very useful and popular:

1. On-demand service – You can use cloud services whenever you need them, just like switching on electricity. No need to set up hardware/software yourself.
2. Broad network access – You can use cloud services from anywhere using the internet (laptop, mobile, tablet).
3. Resource pooling – Cloud providers share their servers and resources among many users. It's like many people using water from the same water tank.
4. Scalability & Elasticity – You can increase or decrease resources (storage, RAM, servers) anytime as per your need. Like stretching or shrinking a rubber band.

5. Measured service (Pay-as-you-go) – You pay only for what you use, just like electricity or mobile data.
6. Security – Cloud providers use encryption, firewalls, and access controls to protect your data.
7. Automatic updates & maintenance – No need to manually update software; cloud services are automatically updated by the provider.
8. High availability & reliability – Cloud is always available (24x7). Even if one server fails, another takes over.

Advantages of Cloud Computing:

1. Cost saving – No need to buy costly servers/software, just pay for what you use.
2. Flexibility & Scalability – Can increase/decrease resources anytime as per need.
3. Accessibility – Access from anywhere, anytime using internet.
4. Data backup & recovery – Cloud keeps data safe and provides easy recovery.
5. Automatic updates – Software and services are updated by provider.
6. Collaboration – Multiple people can work together on the same project easily.
7. Reliability – High uptime; services are available 24x7.

Disadvantages of Cloud Computing:

1. Internet dependency – Cloud works only if internet is available.
2. Security concerns – Data is stored on third-party servers; risk of hacking or misuse.
3. Downtime – Cloud servers may sometimes go down (though rare).
4. Limited control – Users have less control over hardware/software compared to owning it.
5. Hidden costs – Long-term use or extra services may become expensive.
6. Data transfer speed – Uploading/downloading large data may be slow.

LAYERS OF CLOUD COMPUTING

Cloud computing is built on different layers, just like a cake has multiple layers stacked together. Each layer has its own role, but all of them work together to provide smooth

cloud services. The main layers of cloud computing are Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS).

1. Infrastructure as a Service (IaaS)

This is the base layer of cloud computing. In simple terms, it provides the hardware and networking resources we need, but in a virtual form. Instead of buying physical servers, storage devices, or network equipment, we can rent them from a cloud provider. The user gets full control to install any operating system, software, or tools on this infrastructure.

IaaS is a cloud computing model that provides on-demand servers, storage, and networking without the need for physical hardware.

IaaS delivers virtualized computing resources hosted in cloud provider data centers and accessed through a web interface or API.

Businesses can scale resources as needed and pay only for what they use, reducing costs and improving flexibility.

IaaS enables rapid application deployment by eliminating the need for hardware management and simplifying IT infrastructure.

Benefits include reduced costs, flexible scaling, and faster deployment.

IaaS is used for migration, development, storage, web apps, high-performance computing, and big data analysis.

IaaS is evolving with trends like AI-driven automation, edge computing, and eco-friendly infrastructure

IaaS, or Infrastructure as a Service, is a cloud computing model that provides virtualized computing resources over the internet. It offers on-demand access to servers, storage, networking, and other infrastructure components, allowing businesses to rent these resources instead of purchasing and managing their own physical hardware. This enables greater flexibility, scalability, and cost-effectiveness for organizations needing to run applications and store data.

Example: Amazon EC2, Google Compute Engine, Microsoft Azure Virtual Machines.

Example: IaaS = renting the hardware + storage + networking from the cloud.

2. Platform as a Service (PaaS)

This is the middle layer. Imagine you already have the hardware (IaaS), but you also need tools to develop and run applications. PaaS provides you with a ready-made platform where you can build, test, and deploy your applications without worrying

about the underlying infrastructure. It includes operating systems, databases, programming frameworks, and runtime environments

PaaS is a cloud computing model that provides developers with a platform to build, deploy, and manage applications.

It simplifies app development by abstracting infrastructure complexities, allowing developers to focus on building and innovating.

Its Benefits include reduced coding time.

Platform as a service (PaaS) is a cloud computing model that provides developers with a platform to build, deploy, and manage applications without worrying about the underlying infrastructure. It allows developers to focus on writing code, while the cloud provider handles the infrastructure, maintenance, and scalability.

PaaS includes everything needed for application development, including operating systems, runtime environments, databases, development tools, middleware, and hosting and scaling capabilities

Example: Google App Engine, Microsoft Azure App Service, AWS Elastic Beanstalk.

Example: PaaS = renting a development platform (you just focus on coding, not setup).

3. Software as a Service (SaaS)

This is the top layer of cloud computing. It provides ready-to-use applications directly to end users over the internet. Users don't need to install or maintain the software on their own systems. Everything is managed by the cloud provider including updates, security, and performance. We just need a web browser or app to use these services.

SaaS is a cloud-based software delivery model where people access applications over the internet, while the cloud service provider handles infrastructure, security, and updates.

SaaS operates on a subscription basis, eliminating the need for customers to install or maintain software locally.

SaaS applications are highly scalable, allowing customers to adjust their subscription levels as their needs evolve.

Multitenant architecture allows a single instance of SaaS software to serve multiple customers, optimizing resources and reducing costs.

SaaS advantages include cost efficiency, ease of access, and workforce mobility by providing access to apps and data from any internet-connected device.

Example: Google Workspace (Gmail, Docs, Sheets), Microsoft Office 365, Dropbox, Zoom.

Example: SaaS = renting complete software ready to use.

TYPES OF CLOUD COMPUTING:

Cloud computing is not the same for every organization. Different companies and users need different levels of control, security, and accessibility. Based on how the cloud resources are owned, managed, and accessed, cloud computing is divided mainly into three types: Public Cloud, Private Cloud, and Hybrid Cloud. Each type has its own purpose, benefits, and limitations.

1. Public Cloud

The Public Cloud is like a shared playground where everyone can come and use the facilities, but the playground is managed by someone else. In the same way, in public cloud, the cloud services (like storage, computing power, or applications) are provided over the internet by third-party providers such as Amazon Web Services (AWS), Microsoft Azure, or Google Cloud. Many users and companies share the same infrastructure, but their data and applications are kept separate for security.

It is owned and managed by cloud service providers.

Users pay as per usage (like electricity or water bills).

Examples: Gmail, Google Drive, AWS, Azure, Dropbox.

Best suited for startups, small businesses, and individuals who do not want to spend money on their own IT infrastructure.

Advantages of Public Cloud

1. Public cloud is easily able to scale up and down resources as per the demand of traffic and workload. It facilitates with performance optimization and cost efficiency.
2. It works on pay-as-you-go cloud model and helps in resolving the investments needs in hardware and infrastructure reducing overall costs.

Disadvantages of using Public Cloud

1. It is difficult to trust and maintain data to a third-party provider may raise concerns about control and ownership
2. The shared infrastructure of public cloud resources increases the risk of data breaches and unauthorized access. It raises security and privacy concerns.
3. Public clouds come with limited transparency about the underlying infrastructure which may make it challenging to monitor and manage performance effectively.

2. Private Cloud

The Private Cloud is like a personal playground in your own house where only you and your family can play. In the same way, in a private cloud, the cloud infrastructure is exclusively used by a single organization. It can be hosted internally in the company's data center or externally by a third-party provider, but it is dedicated only to that one organization.

Provides high security and privacy because resources are not shared with others. Gives better control over hardware, software, and policies.

Cost is higher compared to public cloud because the organization has to maintain or pay for dedicated infrastructure.

Best suited for banks, government organizations, or large enterprises that handle sensitive data and cannot compromise on security.

Private clouds are distributed systems that work on private infrastructure and provide the users with dynamic provisioning of computing resources. Instead of a pay-as-you-go model in private clouds, there could be other schemes that manage the usage of the cloud and proportionally billing of the different departments or sections of an enterprise. Private cloud providers are HP Data Centers, Ubuntu, Elastic-Private cloud, Microsoft, etc.

Advantages of Private Cloud

1. Better security - All data stays inside the company's own cloud, so chances of leaks are very low.
2. More control - The company fully controls the servers, applications, and policies.
3. Reliable services - Features like data backup, monitoring, clustering, and disaster recovery ensure smooth working.
4. Follow rules & compliance - Companies can set their own rules and meet industry standards (like banking, healthcare).
5. Custom setup - Organizations can design the cloud as per their specific needs.

Disadvantages of Private Cloud

1. Limited access area – It can only be used within the organization, not publicly available everywhere
2. High cost – Setting up private infrastructure is expensive compared to public cloud.
3. Needs skilled experts – Special IT professionals are required to manage, monitor, and maintain it.
4. Scalability issues – Expanding storage and resources may take more time and money.
5. Maintenance responsibility – The organization itself is responsible for upgrades, security patches, and monitoring

3 Hybrid Cloud

A hybrid cloud is a combination of private and public clouds. Organizations use both, depending on the type of data and applications. For example, sensitive data can be stored in a private cloud, while less critical data or applications can run in the public cloud.

Hybrid clouds give the best of both worlds.

It helps organizations maintain security for important data while enjoying the flexibility and cost-saving features of the public cloud.

A hybrid cloud is a heterogeneous distributed system formed by combining facilities of the public cloud and private cloud. For this reason, they are also called heterogeneous clouds.

Benefits of Hybrid Cloud

1. Cost-effective – Cheaper than other cloud types because it uses a distributed system.
2. Fast performance – Reduces latency (delay) in data transfer, making work faster and smoother.
3. Better security – More secure since it works on a distributed system network.
4. Flexibility – Allows combining private (secure) and public (scalable) clouds as per business needs.
5. Scalability – Extra resources from the public cloud can be used during peak demand.

Disadvantages of Hybrid Cloud

1. Need for expertise – Businesses may not have skilled staff to design and manage hybrid environments.
2. Complex security management – Different access rules and security levels for private and public parts can make security harder to manage.
3. Management complexity – Handling both private + public cloud is more complicated than using only one type.
4. Migration issues – Moving data and apps between private and public clouds can be challenging.
5. High setup cost – Though cheaper in the long run, initial setup and integration require higher investment.

1. Serverless Computing

Serverless computing is a cloud computing model where developers do not need to manage servers or infrastructure. They just write code, upload it to the cloud, and the cloud provider takes care of running, scaling, and maintaining it. You are charged only for

the actual execution time, not for idle server time.

Example: AWS Lambda, Azure Functions, Google Cloud Functions.

Uses:

- * Running microservices.
- * Automating tasks and backend processes.
- * Building event-driven applications.
- * Processing API requests and webhooks.

Advantages:

- * No server management required.
- * Automatic scaling based on demand.
- * Cost-effective because you pay only for what you use.
- * Faster deployment and updates.

Disadvantages:

- * Limited execution time per function.
- * Harder to debug and monitor.
- * Dependency on cloud provider's infrastructure.
- * Not suitable for long-running processes.

2. Edge Computing

Edge computing is a distributed computing model where data is processed near the source (the "edge") instead of sending it to a central cloud or data center. This reduces

latency, improves speed, and saves bandwidth.

Example: Smart traffic lights processing data locally, autonomous vehicles analyzing sensor data in real-time, industrial IoT devices monitoring machines.

Uses:

- * Real-time analytics in factories or smart cities.

- * Autonomous vehicles and drones.
- * Streaming and gaming applications.
- * Reducing cloud data transfer costs.

Advantages:

- * Faster response time for real-time applications.
- * Reduces bandwidth usage.
- * Improves reliability during network outages.
- * Better privacy since sensitive data can be processed locally.

Disadvantages:

- * Hardware costs at edge locations.
- * Complex deployment and maintenance.
- * Security risks if local devices are compromised.
- * Limited computing power compared to centralized cloud.

Module 2 – Cloud Security

Shared Responsibility Model

The Shared Responsibility Model explains how security responsibilities are divided between the Cloud Service Provider (CSP) and the customer. The cloud provider is responsible for securing the underlying cloud infrastructure, while the customer remains responsible for securing what they deploy and configure in the cloud.

The exact responsibility changes depending on the service model. In IaaS, the customer has more security responsibility. In PaaS, the provider manages more of the underlying infrastructure. In SaaS, the provider manages most of the technical infrastructure, but the customer still controls things such as users, access, data, and configuration.

The key idea is:

Cloud Provider secures the cloud → Customer secures what they put/configure in the cloud.

This is one of the most important cloud concepts for GRC because **moving to the cloud does not transfer all security responsibility to the provider.**

During a GRC assessment, you need to determine:

- Who is responsible for each control?
- Is the responsibility documented?

- Has the customer configured the service correctly?
- What does the cloud contract/SLA say?
- What security evidence does the provider provide?

Cloud Risks

Cloud risks are security, operational, compliance, privacy, and business risks that can arise from using cloud services. Cloud does not automatically mean insecure, but it changes where systems operate and how security is managed.

Common cloud risks include misconfiguration, excessive permissions, exposed storage, insecure APIs, credential compromise, insufficient logging, data leakage, third-party dependency, compliance issues, and service outages.

A key point is that many cloud incidents are not caused by the cloud technology itself but by incorrect configuration or weak governance.

For GRC, cloud risk assessment is essential because organizations may store highly sensitive information and critical workloads with third-party providers.

You need to evaluate:

- Data sensitivity
- Provider security
- IAM
- Configuration
- Compliance
- Availability
- Third-party risk
- Incident response
- Data location
- Contractual obligations

Common cloud risks:

- **Misconfiguration** → Incorrect security settings.
- **Excessive permissions** → Users/services receive unnecessary access.
- **Data exposure** → Sensitive information becomes publicly or improperly accessible.
- **Credential compromise** → Attackers obtain cloud identities.
- **Insecure APIs** → Weak interfaces expose services or data.
- **Insufficient logging** → Security incidents become difficult to detect/investigate.
- **Third-party dependency** → Business depends on the provider's availability and security.
- **Compliance risk** → Cloud usage may conflict with regulatory or contractual requirements.
- **Availability risk** → Provider outages can affect business operations.

Security Groups

A Security Group is a virtual network security control used in cloud environments to control which network traffic is allowed to reach or leave a cloud resource. You can think of it as a virtual firewall attached to a cloud resource or network interface.

Security groups typically use rules based on things such as source/destination, protocol, and port. They are commonly used to control access to virtual machines, databases, and other cloud resources.

The important idea is:

Resource → Security Group → Allowed/Blocked Network Traffic

Security groups are usually stateful, meaning that when an allowed connection is established, the return traffic is automatically permitted according to the platform's behavior.

Security groups help implement network access control and least privilege in cloud environments.

For GRC, you should verify:

- Only required ports are exposed.
- Internet access is not unnecessarily allowed.
- Administrative ports are restricted.
- Rules have business justification.
- Rules are periodically reviewed.
- Changes are authorized.
- Sensitive resources have stricter network controls.

A common serious mistake is allowing administrative services such as SSH or RDP from anywhere on the Internet.

CSPM

CSPM (Cloud Security Posture Management) is a category of security technology used to **continuously assess cloud environments for security misconfigurations, policy violations, and compliance gaps.**

Instead of manually checking thousands of cloud resources, CSPM tools can continuously evaluate configurations against security policies and recognized benchmarks.

For example, a CSPM tool may identify:

- Publicly exposed storage

- Overly permissive security groups
- Unencrypted resources
- Weak IAM configurations
- Missing logging
- Non-compliant configurations

The key concept is:

Cloud Configuration → Continuous Assessment → Finding → Remediation

Cloud environments change rapidly. A configuration that was secure yesterday can become insecure today because someone changed a setting.

For GRC, CSPM helps with:

- Continuous compliance
- Configuration assessment
- Risk identification
- Security baselines
- Compliance monitoring
- Misconfiguration management
- Evidence collection

However, CSPM does not automatically fix every security problem. Findings still need **risk assessment, prioritization, ownership, and remediation**.

CSPM can identify:

- Public resources
- Excessive permissions
- Missing encryption
- Weak security configurations
- Missing logging
- Policy violations
- Compliance deviations

Typical workflow:

Discover → Assess → Identify Finding → Prioritize → Remediate → Verify

CSPM can assess environments against:

- Organizational policies
- Security baselines
- CIS benchmarks
- Regulatory requirements
- Cloud security standards

Compliance

Cloud compliance means ensuring that the organization's cloud usage, configurations, processes, and data handling meet applicable laws, regulations, contractual obligations, and security standards.

Moving systems to a cloud provider does not remove the organization's compliance obligations. The organization must determine which requirements apply and how those requirements are implemented in the cloud environment.

Cloud compliance can involve frameworks and requirements such as ISO/IEC 27001, SOC 2, GDPR, PCI DSS, HIPAA, or industry-specific regulations, depending on the organization.

Cloud environments can create compliance challenges involving:

- Data location
- Privacy
- Access control
- Encryption
- Logging
- Retention
- Third-party risk
- Regulatory requirements
- Evidence availability

For GRC, you must distinguish between **provider compliance** and **customer compliance**.

A cloud provider having an ISO 27001 certificate does not automatically mean your company's cloud environment is ISO 27001 compliant.

Compliance requires understanding:

- Applicable regulations
- Contractual requirements
- Security frameworks
- Data classification
- Data residency
- Access controls
- Encryption
- Logging and monitoring

- Retention requirements
- Third-party assurance

Useful evidence can include:

- Provider certifications
- SOC reports
- Policies
- Configuration evidence
- Access reviews
- Audit logs
- Risk assessments
- Security assessments

Governance

Cloud governance is the system of policies, processes, roles, responsibilities, standards, and controls used to manage cloud environments in a controlled and secure way. It ensures that cloud usage supports business objectives while managing security, risk, compliance, cost, and operational requirements.

Cloud governance answers questions such as who can create cloud resources, what security requirements they must follow, where data can be stored, how access is controlled, what must be logged, and how compliance is monitored.

A useful way to understand it is:

Business Requirements → Cloud Policies → Controls → Monitoring → Review → Improvement

For a GRC professional, cloud governance is much more important than simply knowing how to configure cloud services. Organizations can have technically secure cloud infrastructure and still have poor governance.

You need governance to prevent:

- Unauthorized cloud resources
- Excessive privileges
- Unapproved services
- Data being stored in prohibited locations
- Inconsistent security configurations
- Uncontrolled cloud spending

- Compliance violations
- Unmanaged third-party risk

Cloud governance commonly covers:

- **Identity governance** → Who can access cloud resources?
- **Data governance** → What data can be stored and where?
- **Security governance** → What security controls are mandatory?
- **Compliance governance** → Which requirements must be followed?
- **Configuration governance** → What configurations are allowed?
- **Resource governance** → Who can create or modify resources?
- **Risk governance** → How are cloud risks identified and treated?
- **Cost governance** → How is cloud spending controlled?

Important governance mechanisms include:

- Cloud security policies
- Security baselines
- Standards
- IAM policies
- Approval processes
- Configuration rules
- Continuous monitoring
- Compliance assessments
- Risk registers
- Exception management

Best Practices

Cloud security best practices are proven security approaches used to reduce cloud risk and maintain a secure, controlled environment. They are not a single technology or product. They combine identity security, secure configuration, data protection, monitoring, resilience, vulnerability management, and governance.

The goal is to make security consistent and repeatable, rather than depending on individual administrators making correct decisions every time.

For GRC, best practices provide a practical basis for creating security requirements, control objectives, assessment criteria, and audit checklists.

A cloud environment should not only be secure today; it should remain secure as resources, users, applications, and configurations change.

Important practices include:

- Apply **least privilege**.
- Enable **MFA**, especially for privileged accounts.
- Use centralized IAM.
- Encrypt sensitive data.
- Protect encryption keys.
- Disable unnecessary services.
- Restrict network exposure.
- Avoid publicly exposing sensitive resources.
- Maintain secure configuration baselines.
- Continuously monitor cloud environments.
- Enable appropriate logging.
- Integrate relevant logs with SIEM.
- Regularly assess vulnerabilities.
- Maintain tested backups.
- Establish incident-response procedures.
- Review third-party/cloud-provider risks.
- Conduct periodic access reviews.
- Continuously assess compliance.
- Maintain documented exceptions and risk acceptance.